Integrated Circuit ECE326
Lec02
Synthesis Digital Circuit

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Outlines

- Integrated Circuit (IC)
- Transistor History
- Integrated Circuit (IC) History
- Beyond the end of Moore’s law
- Integrated Circuit Package
- Integrated Circuit Classification

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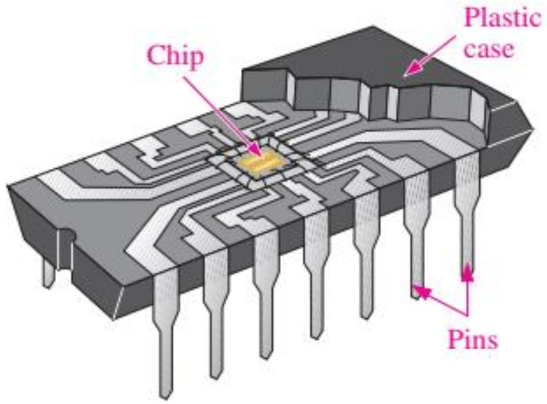
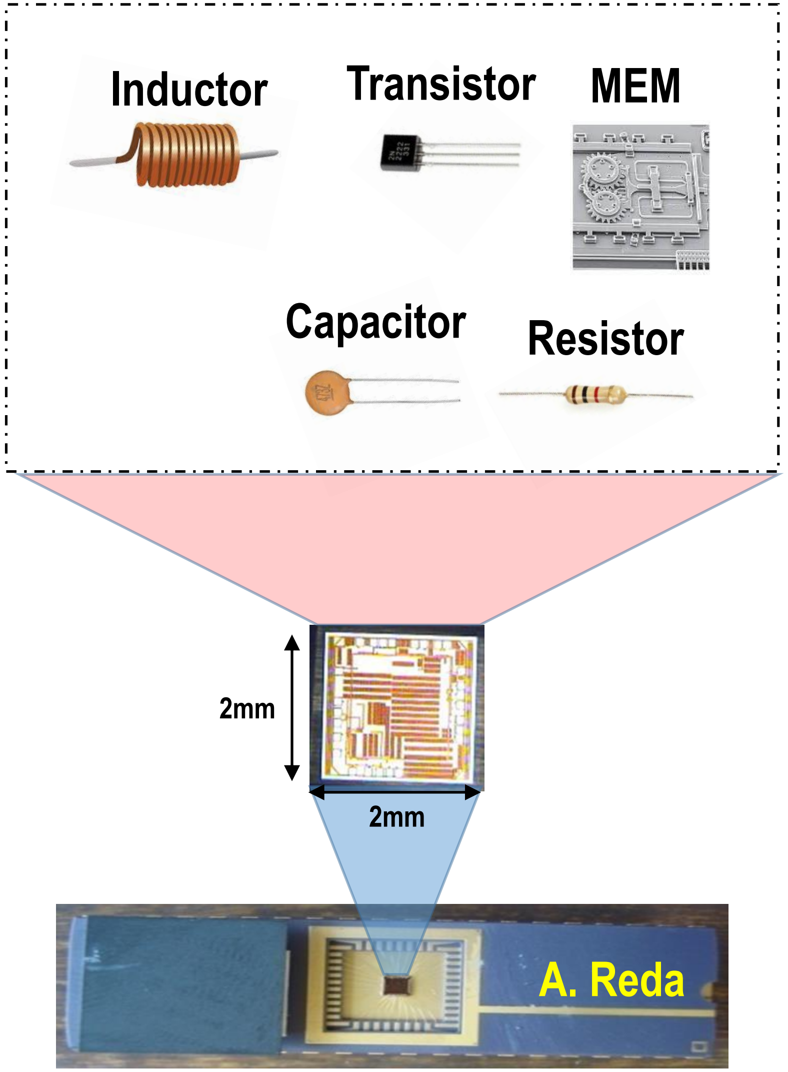
Integrated circuit (IC)

- An **integrated circuit** (IC) is a set of electronic circuits on one small flat piece (or "chip") of semiconductor material, usually silicon.
- **Advantages of IC**
  - Small size
  - High reliability
  - Low cost for mass production
  - Low power consumption



Cutaway view of one type of fixed-function IC package



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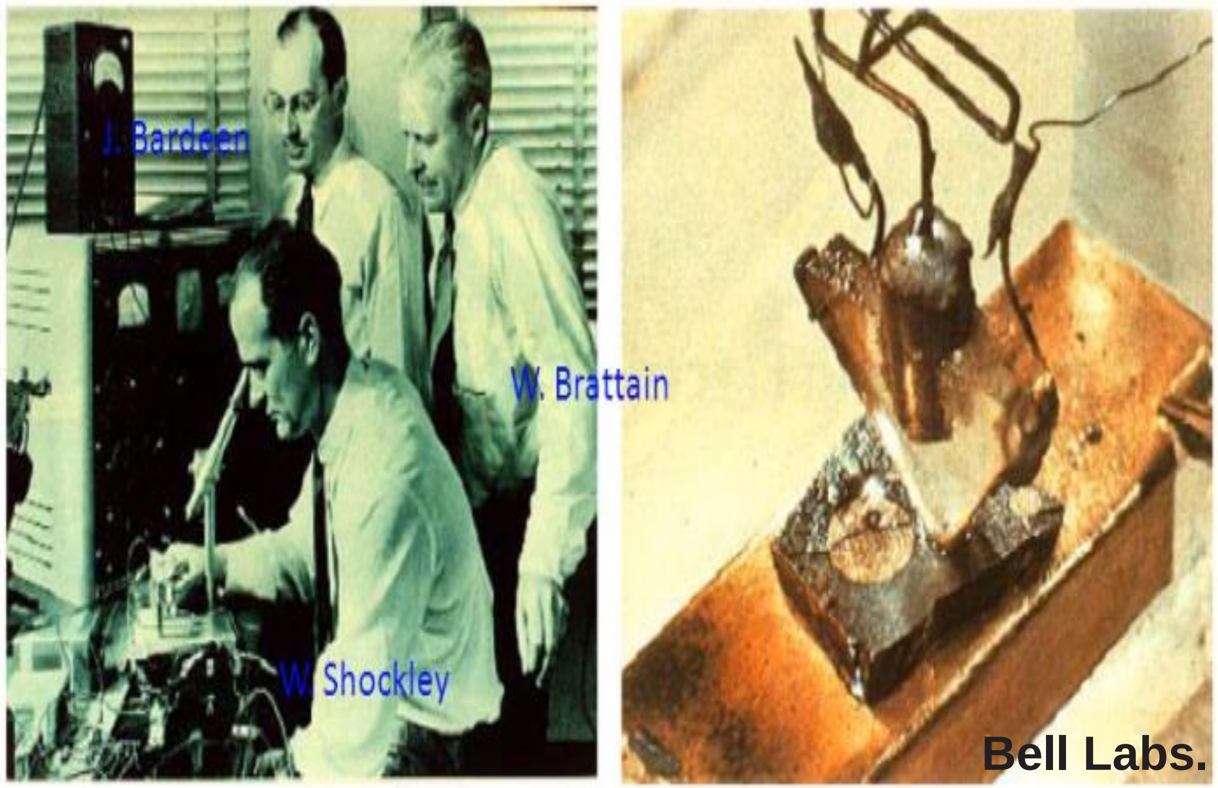
Transistor History (1/2)

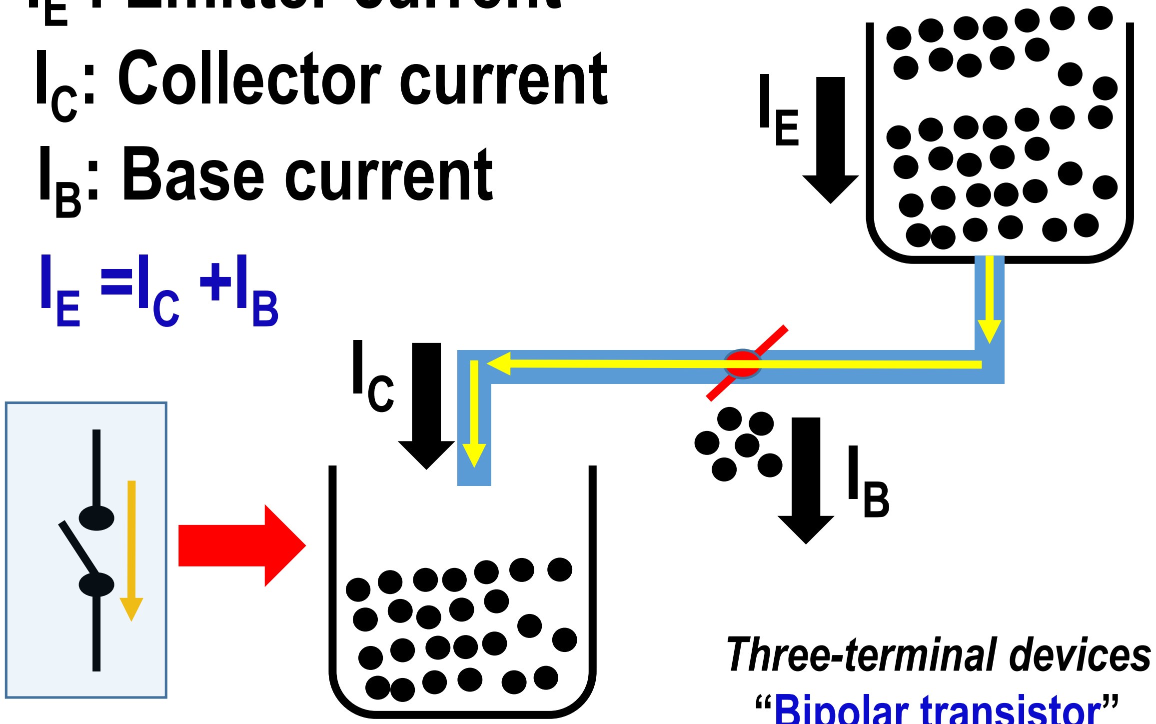


I_E : Emitter current

I_C: Collector current

I_B: Base current

$I_E = I_C + I_B$

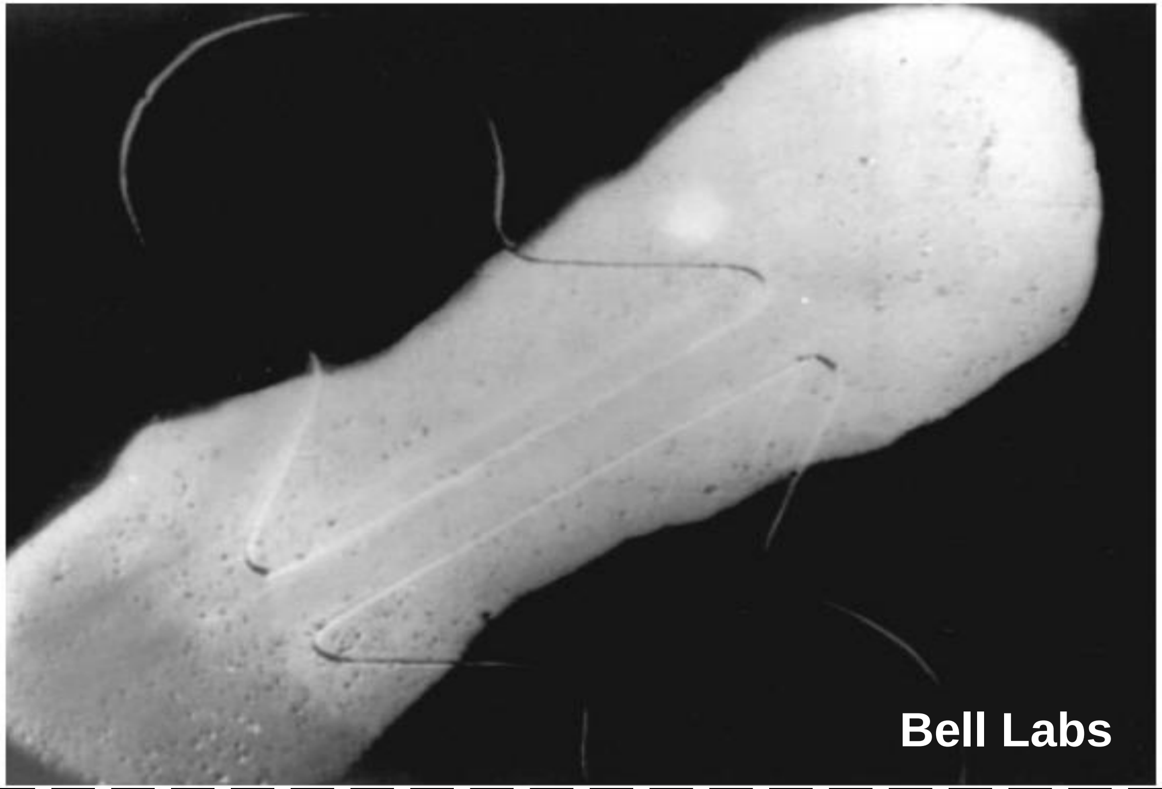


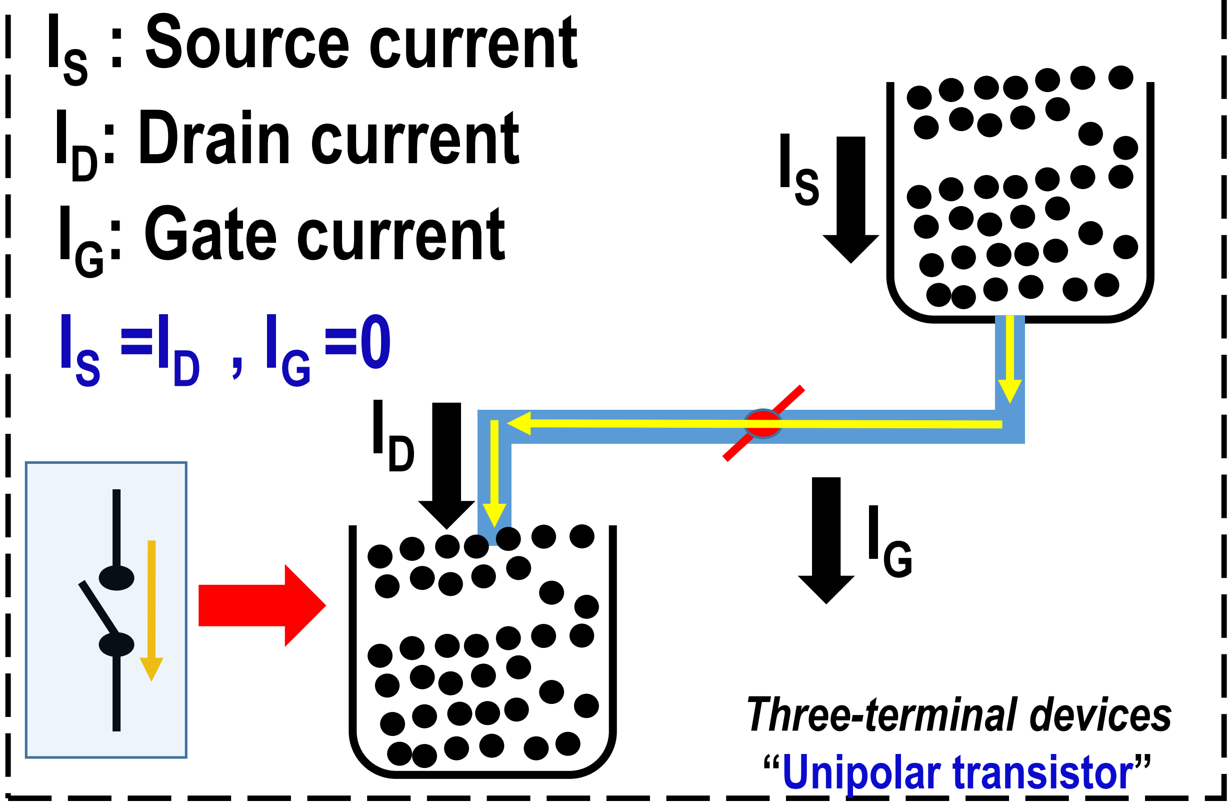
Three-terminal devices

"Bipolar transistor"

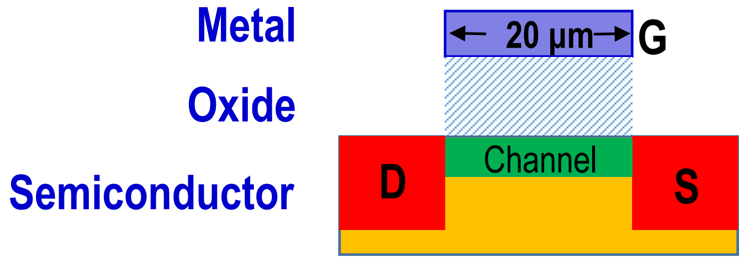
- **In1947, the first solid-state transistor was invented in Bell Labs.**
- **William Shockley, John Bardeen, Walter Brattain (1956 Nobel Prize in Physics)**
- Noted, the two point contact separated by about **50 µm**
- The used semiconductor was **Germanium, it is replaced by Silicon….Why?**
    - **Si's oxide is stable, more economic, less sensitive to temperature**

Transistor History (2/2)





- ❑In1960, Kahng and Atalla was reported a Metal-Oxide-Semiconductor (**MOS**)
- ❑The MOS field-effect transistor (**MOSFET**) is the most important device for advanced integrated circuits → Gate length of **20μm**
- ❑ The MOSFET with a gate length of **5 nm** has been demonstrated → (**one trillion devices/Chip**)



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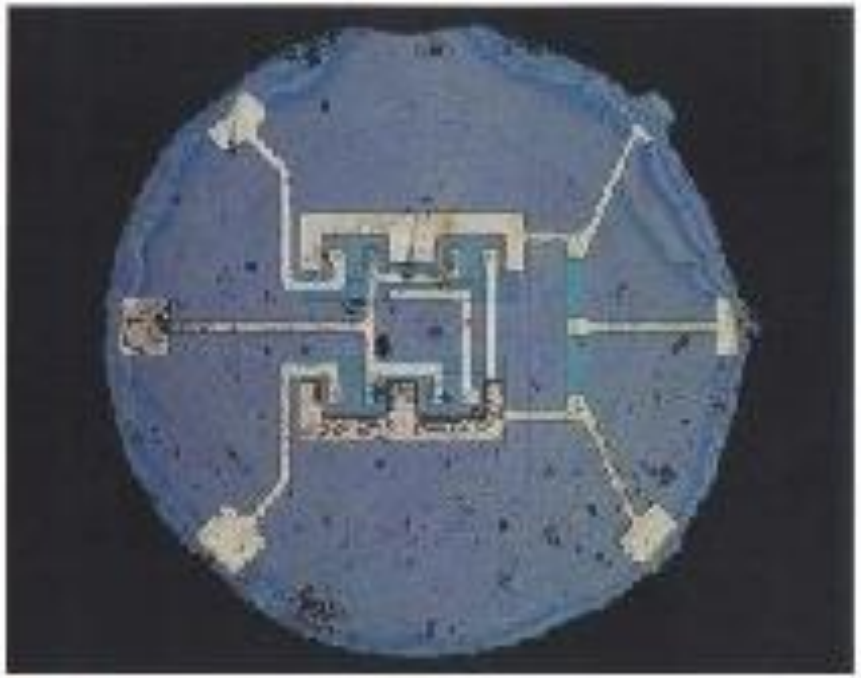
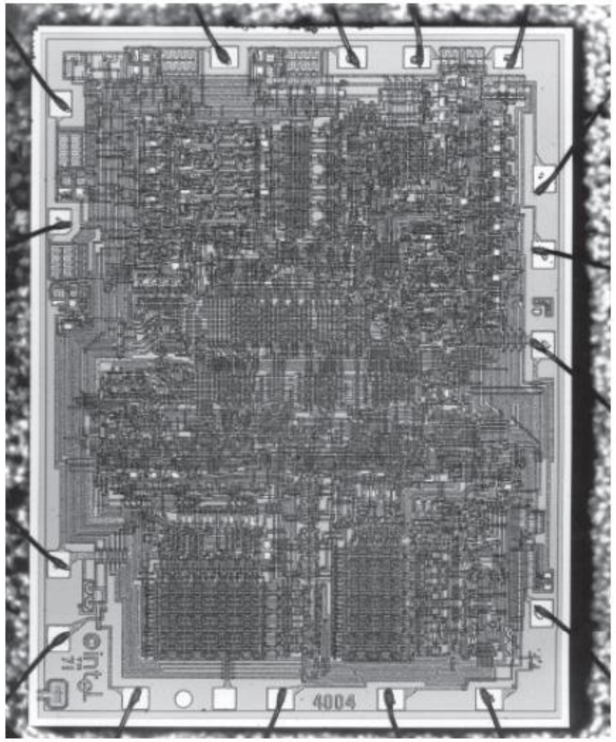
Integrated Circuit History (1/5)

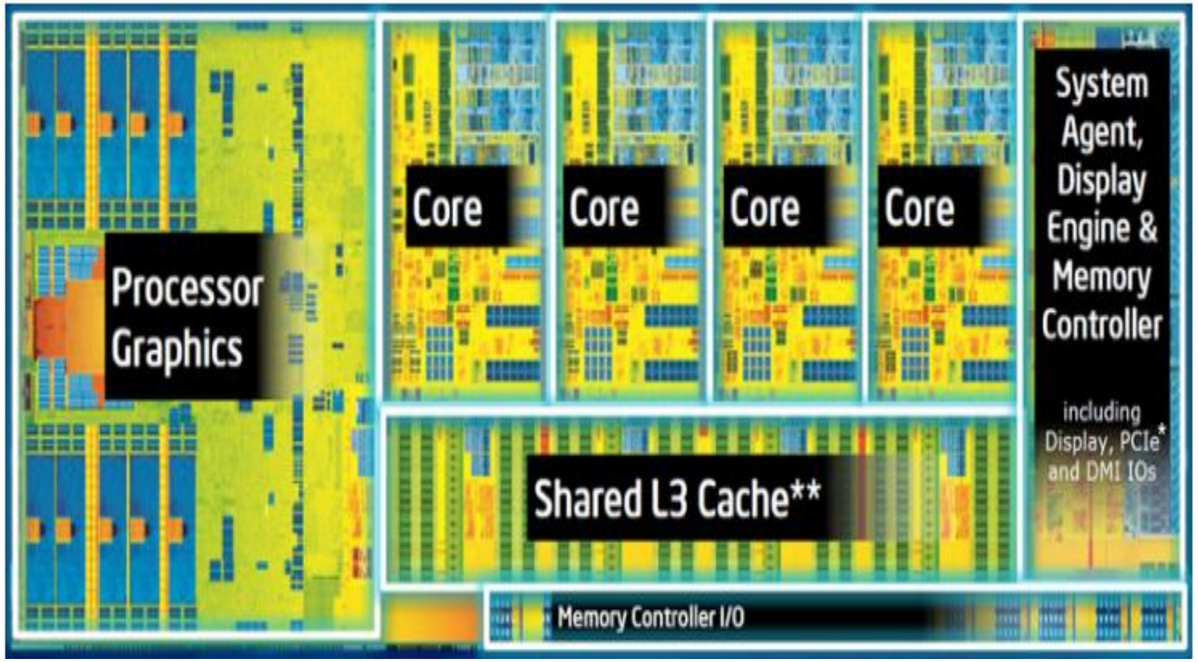


- **In 1959**, a rudimentary IC was invented by **Jack Kilby**
- Taxes instruments, Inc.
- **1X** bipolar transistor, **3X** resistors, and **1X** Capacitor

- **In 1961**, the **first** planar IC was invented by **Robert Noyee**
- Fairchild Semiconductor
- **1X**Transistor,**1X** Resistor,**1X** Capacitor
- **Monolithic IC** (single stone)

Integrated Circuit History (2/5)

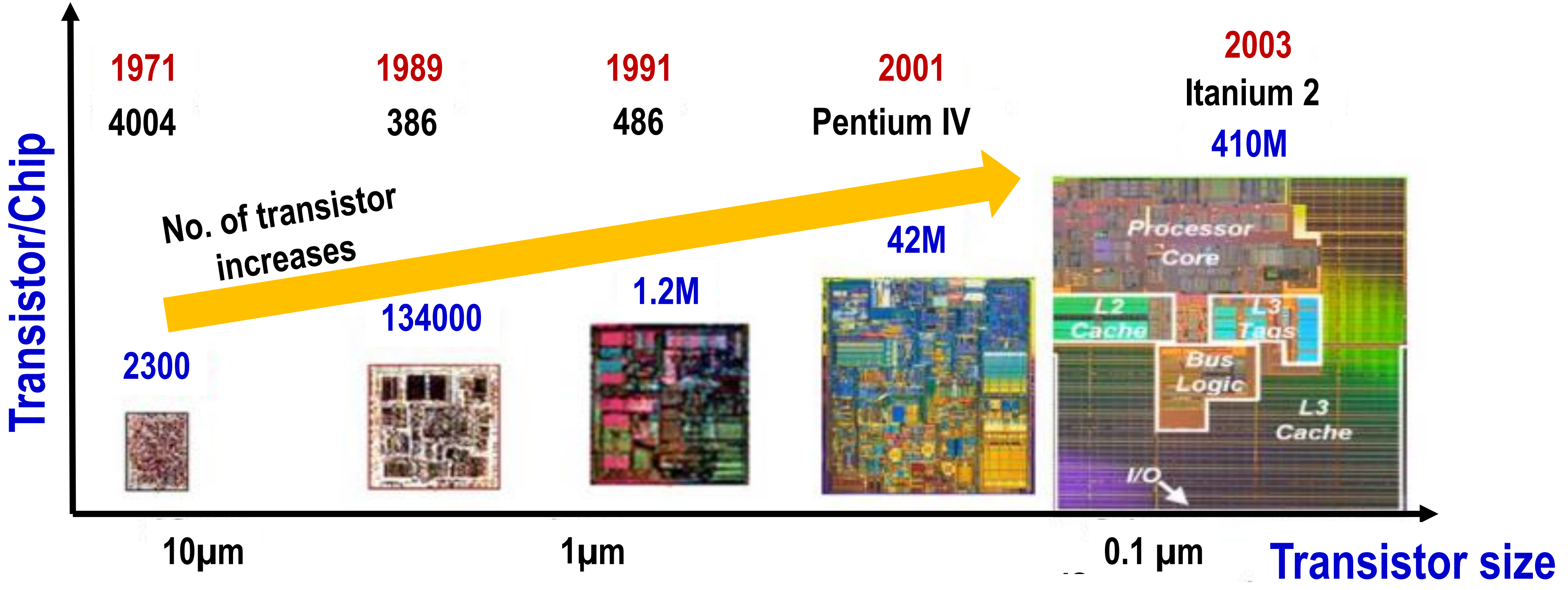




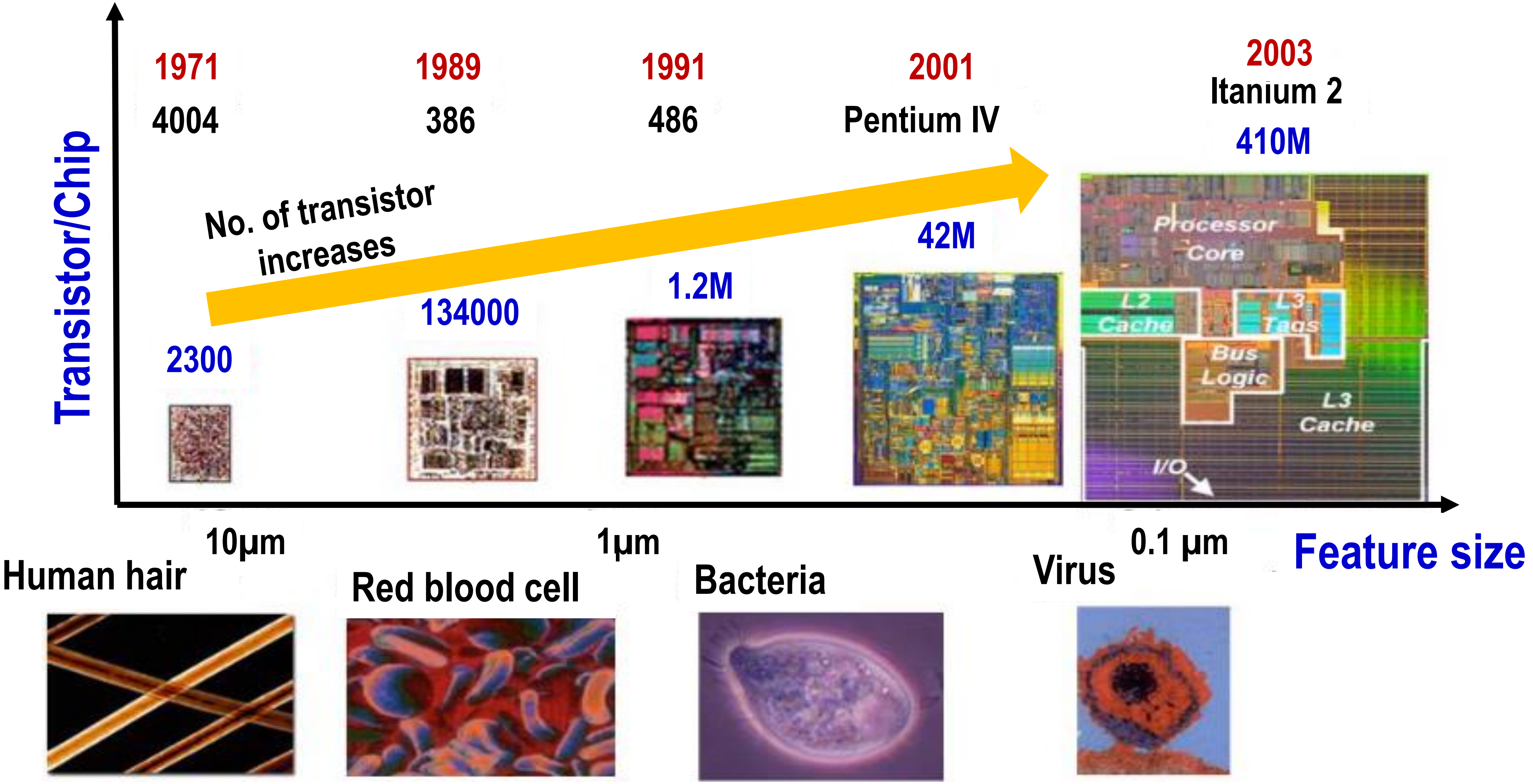
- ❑In **1971**, the first microprocessor was made by **Hoff**
- ❑**4-bit** microprocessor (**Intel 4004**)
- ❑Chip size of **12mm^2**
- ❑It contained **2300** MOSFETs
- ❑Operated at **0.1 MIPS** @Freq. of **470KHz**

- ❑In **2017**, the Quad microprocessor was invented
- ❑**64-bit** microprocessor (**Intel® Core™ i7-4700MQ**)
- ❑Chip size of **174.4mm^2**
- ❑It contained **> Billions** MOSFETs
- ❑Operated at **100,000 MIPS** @ **5GHz**

Integrated Circuit History (3/5)

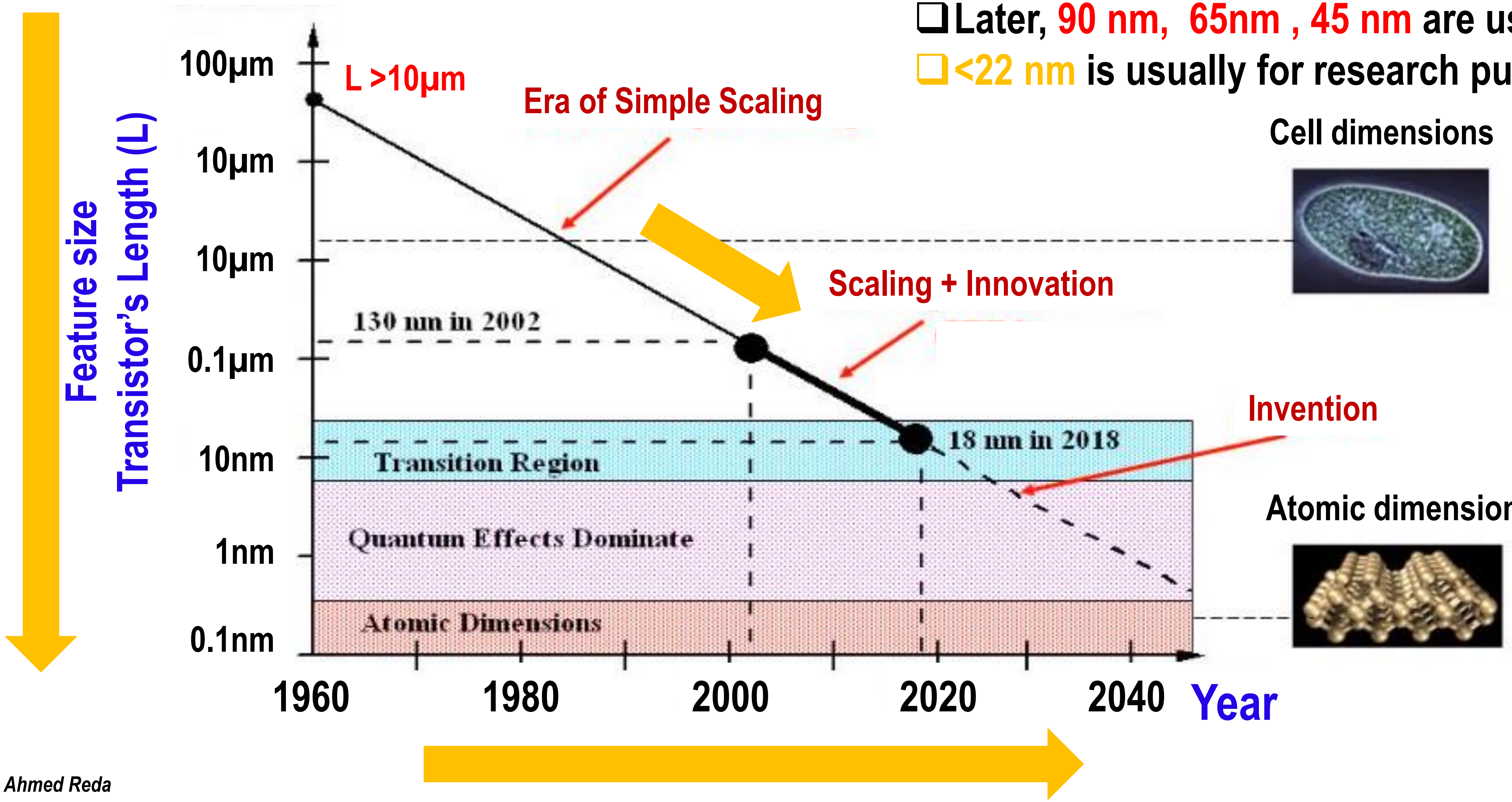


Integrated Circuit History (4/5)



Integrated Circuit History (5/5)

- ❑ 350 nm, 180 nm,130 nm are wieldy used.
- ❑ Later, 90 nm, 65nm , 45 nm are used.
- ❑ <22 nm is usually for research purposes.



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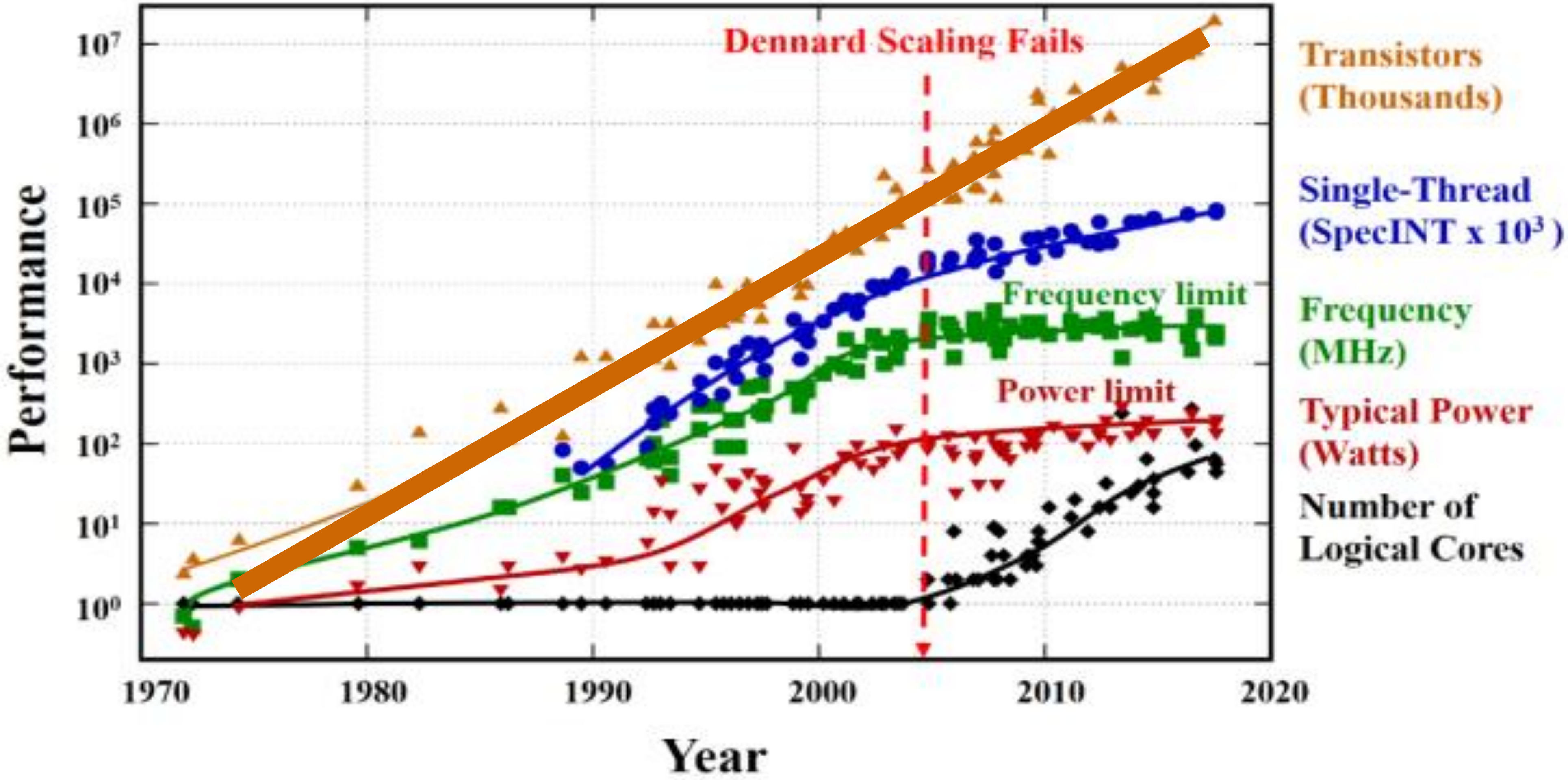
Beyond the end of Moore's law (1/5)

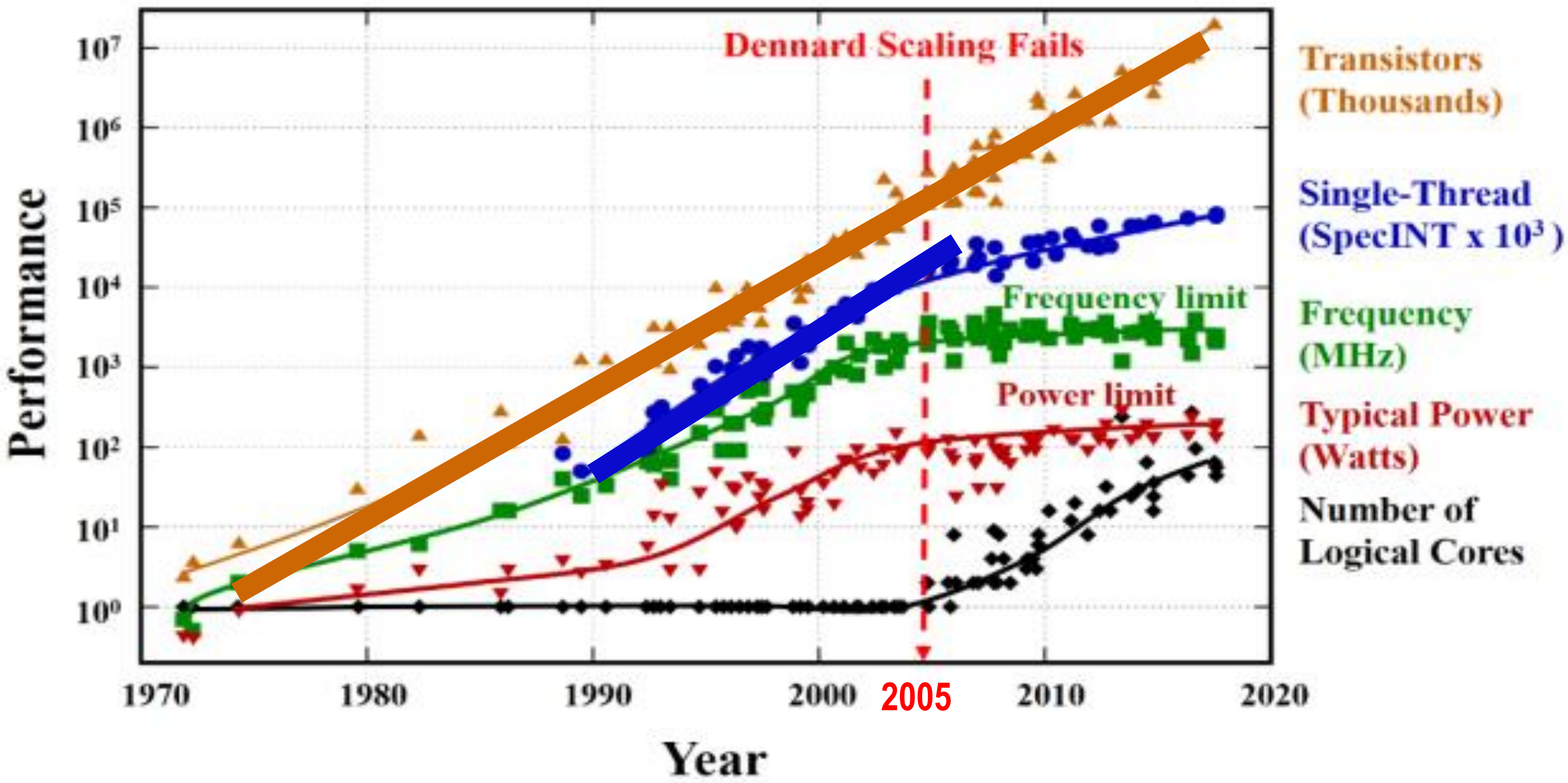
1µm→350nm→180nm→130nm→90nm→65nm→45nm…..→5nm



- **Moore's law** →industrial information technology has increased digital electronics' performance **2X (Doubled) every 2-years within a fixed expense**.

Beyond the end of Moore's law (2/5)

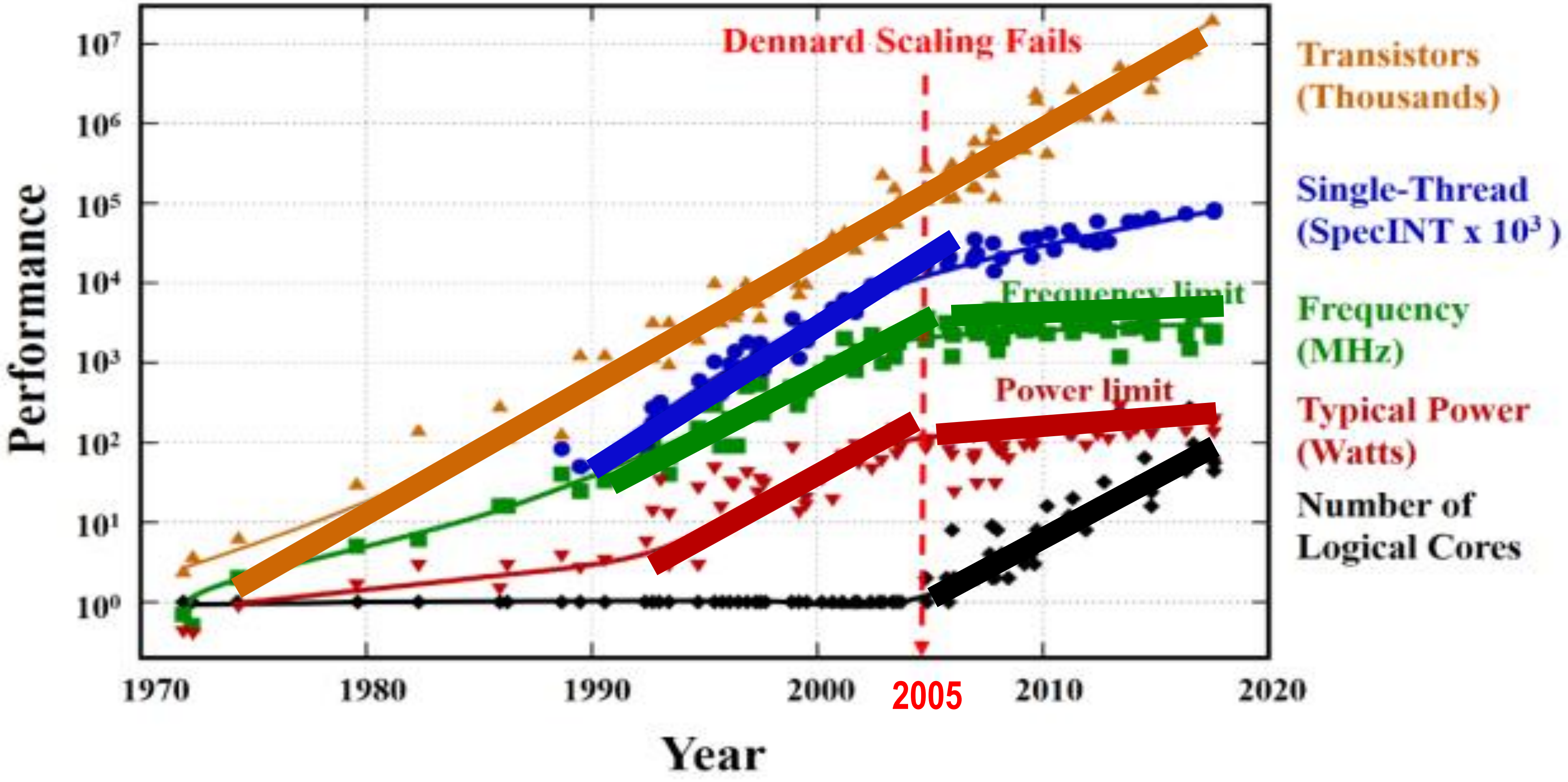
1μm→350nm→180nm→130nm→90nm→65nm→45nm…..→5nm



- ❑ **Scaling down** (**Dennard scaling**) aims to scale the transistor down and miniaturize its to improve the performance

Beyond the end of Moore's law (3/5)

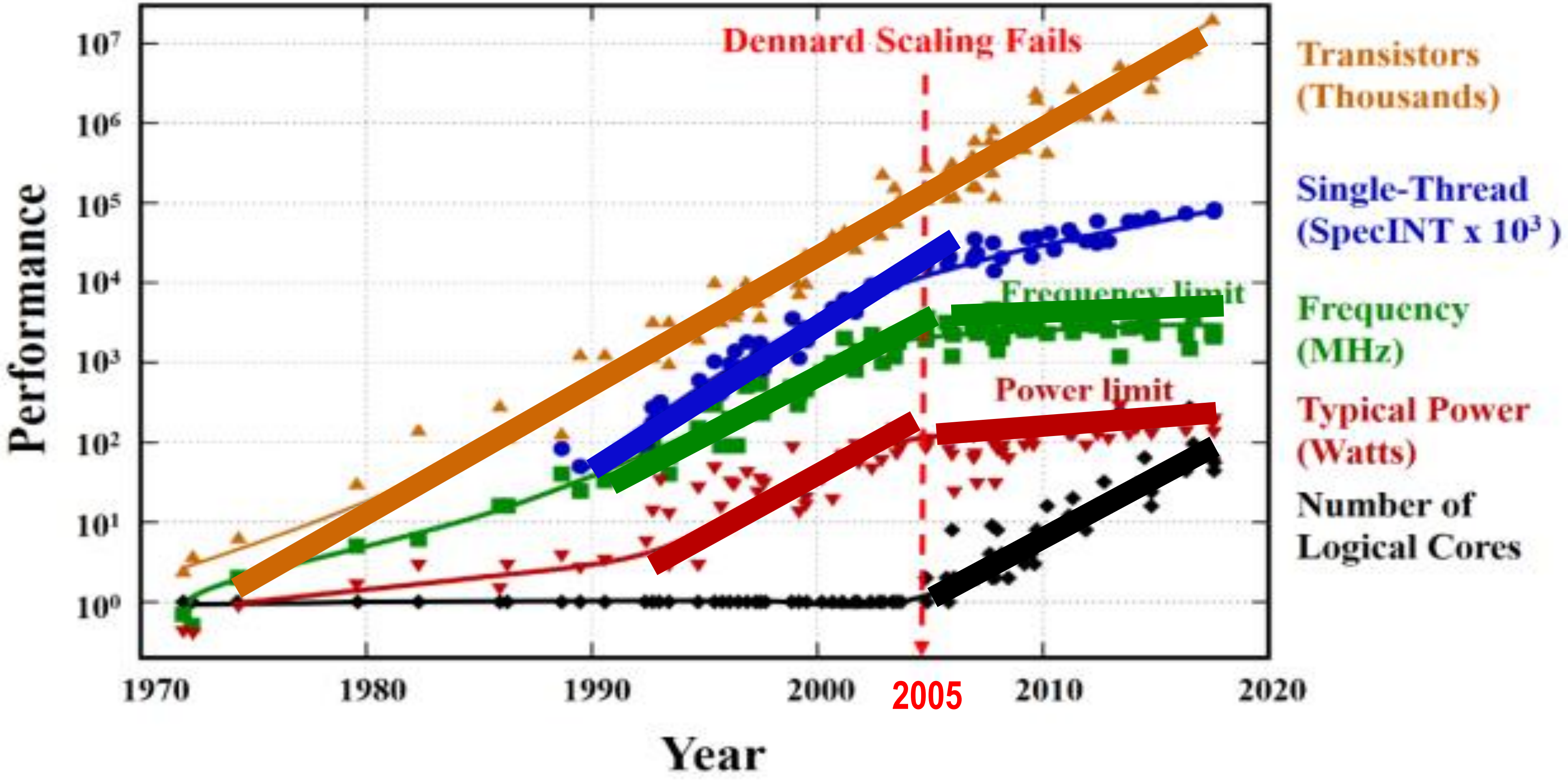
1µm→350nm→180nm→130nm→90nm→65nm→45nm…..→5nm



- ❑ Dennard scaling fails in 2005 due to the heat-removal issue.
- ❑ Multicore architecture → improve the compute capability, keep Moore's law valid from 2005 to 2014

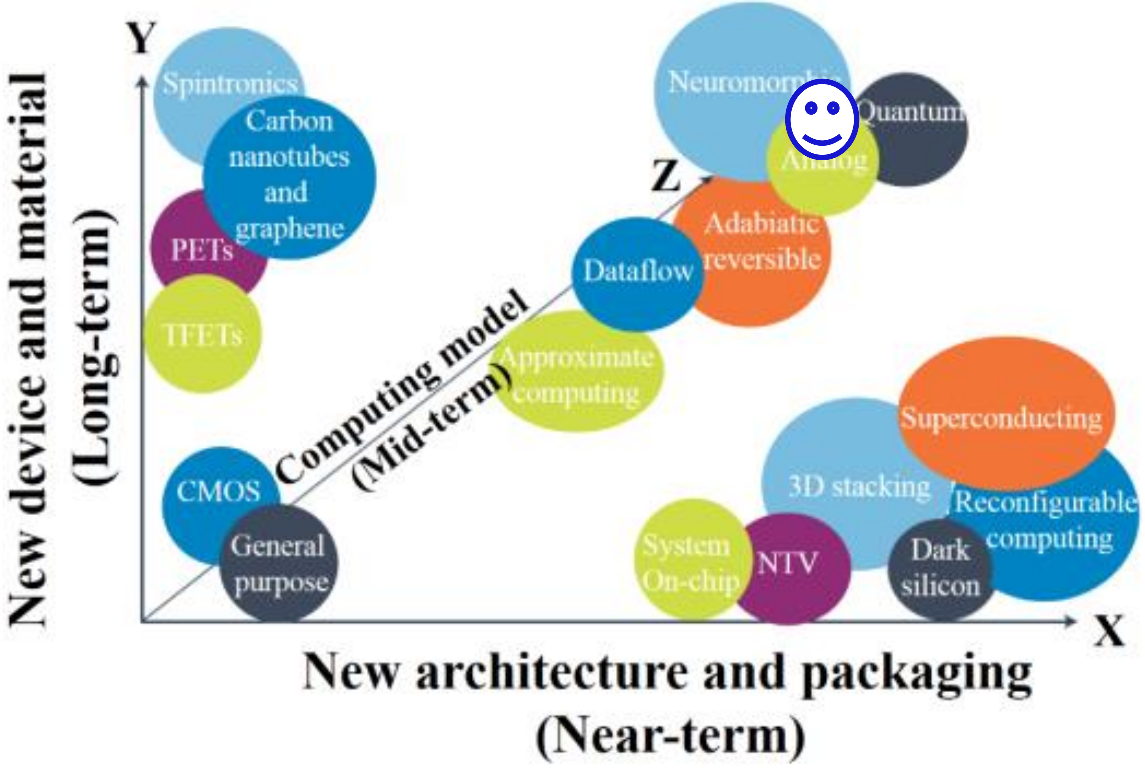
Beyond the end of Moore's law (4/5)

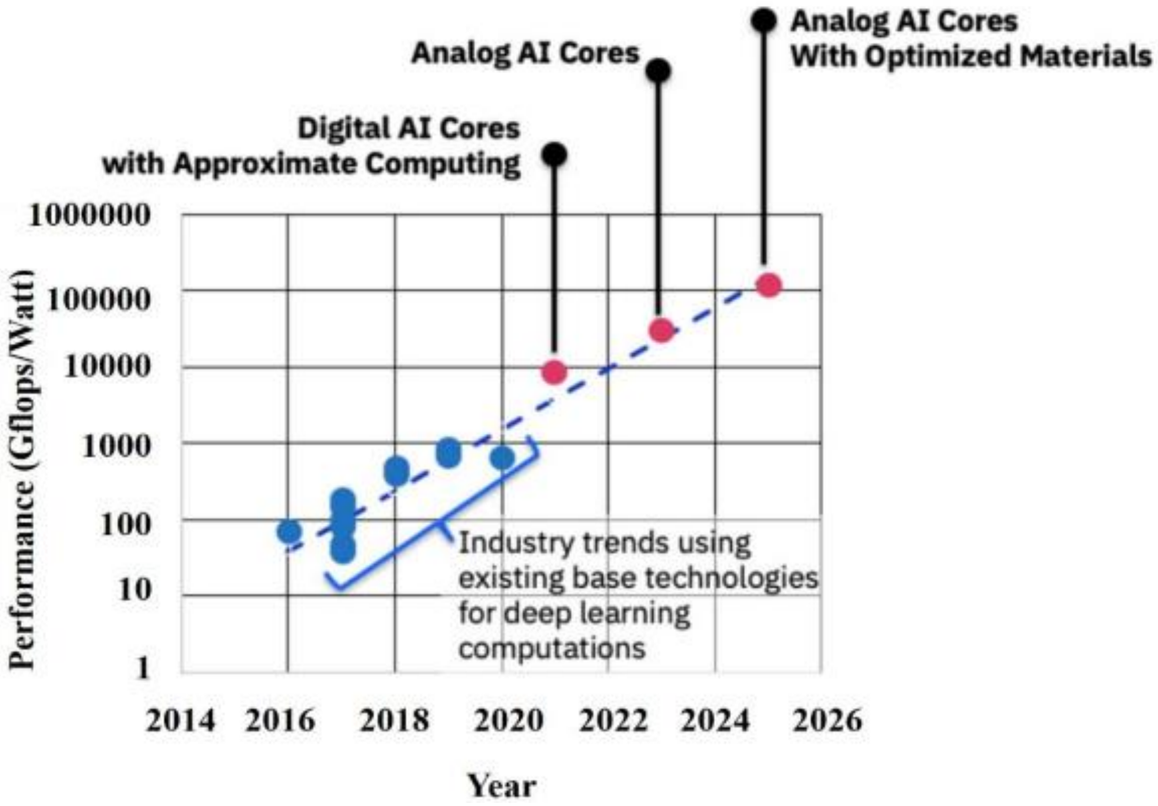
1µm→350nm→180nm→130nm→90nm→65nm→45nm…..→5nm



- However, scaling down the transistor more and more led to hit the brick wall of physics because the transistor would reach atomic-scale, resulting in breaking down Moore's law

Beyond the end of Moore's law (5/5)





- ❑The absence of Moore's law led to seek alternative materials, physical devices, computation models to overcome the technology scaling issue.
- ❑Today, IBM has launched Research Collaboration Center (RCC) to drive next-generation AI hardware.

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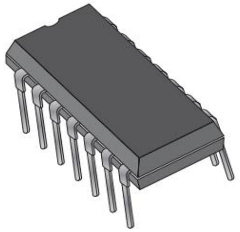
Integrated Circuit Package (1/2)

IC Package

Printed Circuit Boards (PCBs)

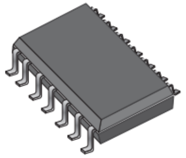
Through-hole mounted



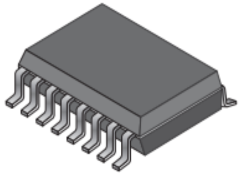
Dual in-line package (DIP)

Surface-mount Technology (SMT)

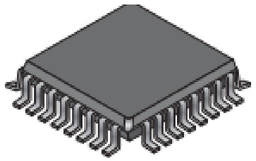
- The holes are unnecessary
- SM package is much smaller



Small-Outline Integrated Circuit (SOIC)



Shrink Small-Outline Package (SSOP)

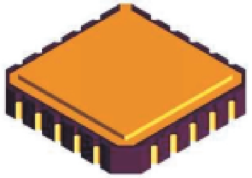


Low-profile Quad Flat Package (LQFP)

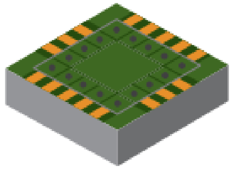
Gull-wing shape

Plastic-Leaded Chip Carrier (PLCC)

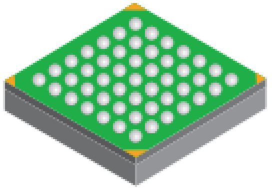
J-type shape



Leadless Ceramic Chip (LCC)



Chip scale package (CSP)



Fine-Pitch Ball Grid Array (FBGA)

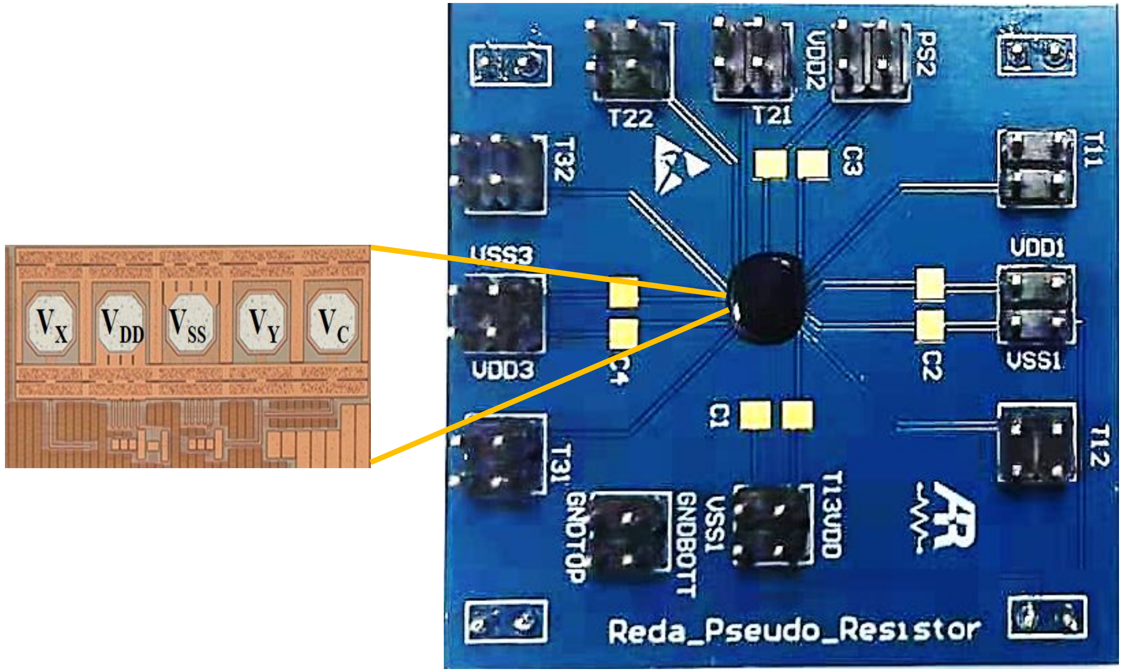
Chip-On-Board

Integrated Circuit Package (2/2)

IC Package

Printed Circuit Boards (PCBs)

Chip-On-Board



A chip-on-board (COB) is a chip that is mounted directly on a circuit board as opposed to being through-hole or surface-mounted technology.

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Integrated Circuit Classification (1/3)

Complexity

Small-scale integration (SSI)

- 10 gates.
- Basic gates and flip-flops.

Medium-scale integration (MSI)

- 10 to 100 gates.
- Encoders, decoders, counters, registers, multiplexers, arithmetic circuits, small memories.

Large-scale integration (LSI)

- 100 to 10,000 gates.
- Memories

Very large-scale integration (VLSI)

- >10,000 to 100,000 gates

Ultra large-scale integration (ULSI)

- > 100,000 gates

Integrated Circuit Classification (2/3)

Technology

MOSFETs
(metal-oxide semiconductor field-effect transistors)

- ❑ Low speed ☹
- ❑ Low power ☺
- ❑ Small area

BJT
(bipolar junction transistors)

- ❑ High speed ☺
- ❑ High power ☹
- ❑ Large area

BiCMOS
uses a combination of both CMOS and bipolar

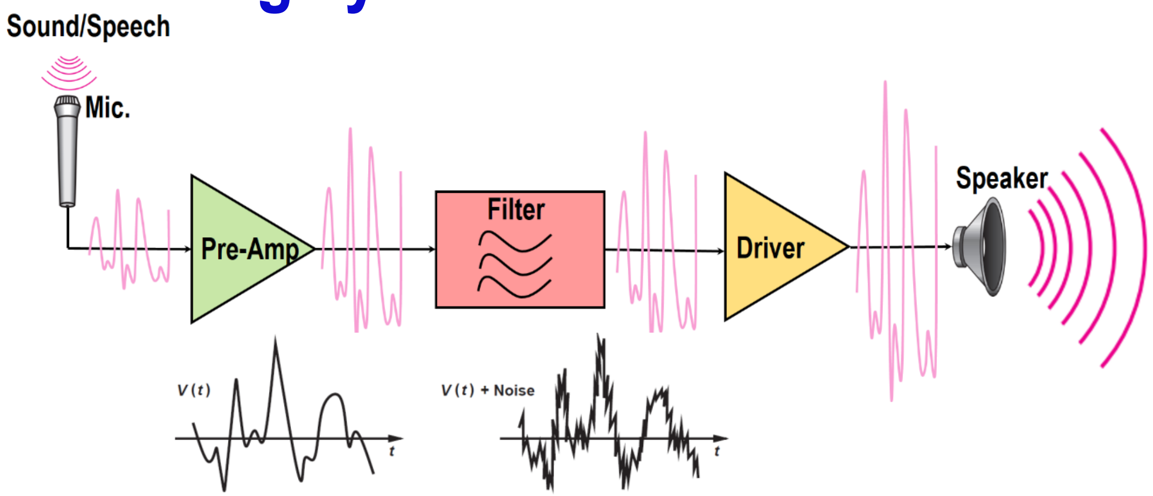
- ❑ High expensive ☹

❑ Advantage of CMOS technology ☺

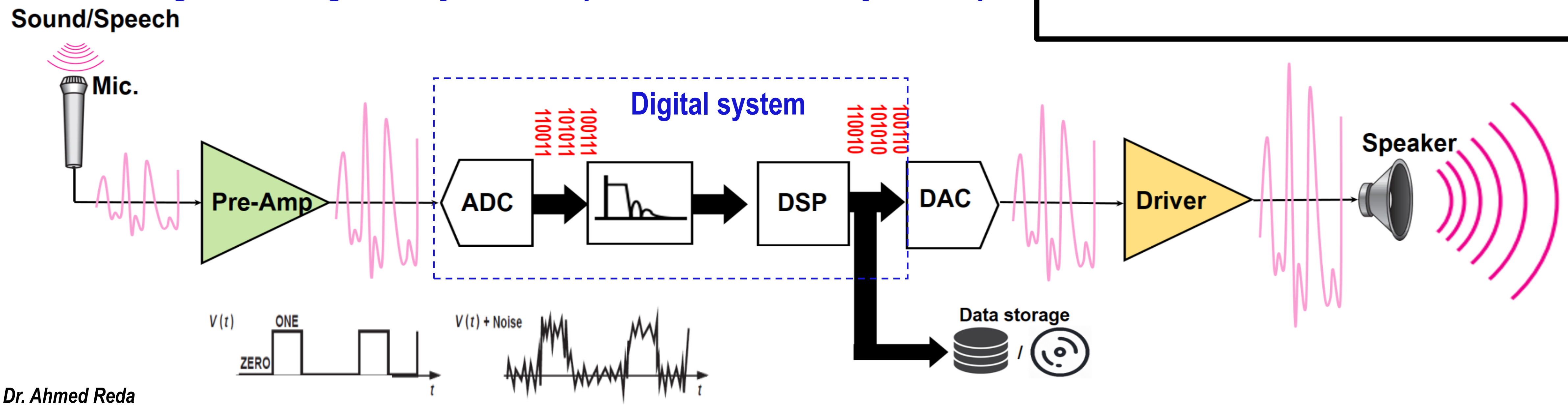
- ❑ Cheapest technology for high density transistor.
- ❑ Memory can be implemented density.
- ❑ Fully restored level.
- ❑ Very low static power.
- ❑ Symmetrical device (symmetrical rise and fall time)

Integrated Circuit Classification (3/3)

- ❑ **Analog system**



- ❑ **Analog and digital system (mixed-mode system)**



Digital System Advantages

High immunity to noise (Robust).
Stored in a digital memories.
Easily compressed and encrypted.

Digital System Disadvantages

Sampling may cause loss of information.
A/D and D/A consume high power.
Processing is more complex.

Thanks!